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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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CERTIFICATE

This is to certify that Project Report entitled **LoanLens** which is submitted by Bhavya Kumar, Abhessht Mishra, Ayush Pratap Singh in partial fulfillment of the requirement for the award of degree B. Tech. in the Department of CSE (AI) of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ABSTRACT

Loan approval is a critical process in financial institutions that requires accurate assessment of an applicant's creditworthiness. **LoanLens** presents a machine learning-based loan classification system designed to predict whether a loan application should be approved or refused based on financial and credit-related features. The dataset includes attributes such as income, credit score, employment history, loan amount, and credit behavior, which are used to train and evaluate multiple models.

A comparative analysis is performed using GBM, XGBoost, and a deep learning model developed with H2O.ai and Scikit-learn. Among these, the deep learning model demonstrates superior performance and is selected as the final prediction model. To address the challenge of interpretability, SHAP is applied using the GBM model to provide feature-level explanations for predictions.

The system is deployed as a RESTful API using FastAPI, enabling real-time predictions based on user-provided input data. The proposed approach combines predictive accuracy with explainability, making it suitable for supporting data-driven decision-making in loan approval systems.

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LIST OF ABBREVIATIONS

API	Application Programming Interface
AUC	Area Under the ROC Curve
DL	Deep Learning
EDA	Exploratory Data Analysis
FastAPI	A modern, web framework for building API
GBM	Gradient Boosting Machine
H2O.ai	An open-source, distributed machine learning platform
ML	Machine Learning
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
SHAP	SHapely Additive exPlanations
XGBoost	Extreme Gradient Boosting

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Loan approval is an essential process in financial institutions, where lenders must evaluate the creditworthiness of applicants before granting loans. Traditional evaluation methods often rely on manual analysis and predefined rules, which can be time-consuming and sometimes inconsistent. With the growth of financial data and advances in machine learning, predictive models can assist institutions in making faster and more accurate lending decisions. Machine learning algorithms are capable of analyzing patterns in historical financial data and identifying factors that influence loan repayment behavior. This project explores the use of machine learning techniques to build a loan classification system that predicts whether a loan application is likely to be approved or rejected. By automating parts of the decision-making process, such systems can improve efficiency, reduce risk, and support data-driven financial analysis.

1.2 PROJECT DESCRIPTION

The focus of this project is on developing a loan classification system that can predict loan approval using machine learning techniques. The system is designed to consider different financial and personal characteristics of loan applicants, such as income levels, credit scores, employment status, loan amount, and credit history. The machine learning algorithm developed through H2O.ai is used to process all this information and generate a probability score for loan approval.

The trained model is served via a REST API created using FastAPI, enabling users or other applications to submit loan applications in JSON format and receive predictions in real-time. Data validation and preprocessing occur before sending it to the trained model for prediction.

After predicting the probability, based on decision thresholds, the final classification is returned to indicate whether to accept or refuse the loan.

In conclusion, this project showcases how machine learning models can be combined with modern API libraries to build efficient predictive systems for decision support in finance.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION

Traditionally, the financial industry has relied on the concept of credit scores in assessing the creditworthiness of individuals and institutions. Statistical techniques, including logistic regression, have traditionally formed the core of financial models in this context. However, the growing complexities and dimensions of financial data have also highlighted the shortcomings of these conventional approaches. In this context, the recent rise of ML and DL models has proven to be far superior in the field of loan prediction models. This literature review attempts to critically evaluate the recent developments in the field of loan prediction models, the shortcomings of the existing literature, and the role of the “Loan Lens” project in addressing these shortcomings.

2.2 TRADITIONAL CREDIT SCORING APPROACHES

Traditional credit scorecards, built primarily using logistic regression methods, have been found to deliver high levels of interpretability and ease of deployment. However, they have been found to be lacking in situations where non-linear relationships or large amounts of multidimensional data are encountered. This has created a requirement for more sophisticated methods that can handle complex data relationships without sacrificing interpretability.

2.3 MACHINE LEARNING AND ENSEMBLE METHODS

Machine learning has significantly improved the predictive capabilities of credit risk models by leveraging algorithms adept at identifying non-obvious data patterns. Ensemble learning techniques, such as XGBoost, have garnered attention for their predictive power and efficiency. For instance, Nurgoho (2024) demonstrated a personal loan prediction accuracy of 93.4% using an XGBoost-based model, surpassing the performance of traditional approaches.

Model interpretability, however, remains a critical concern. To address this, SHapley Additive explanations (SHAP) has been widely adopted, offering detailed insights into the contribution of individual features to specific predictions. Despite its strengths, some studies argue that

SHAP may not consistently outperform simpler feature selection methods across all contexts, suggesting the need for further comparative analysis.

2.4 DEEP LEARNING AND OPEN BANKING DATA

The advent of open banking APIs and the growing availability of granular transactional data have facilitated the incorporation of text-based and sequential inputs into loan

prediction models. A recent study published in MDPI (2024) compared a deep learning model built from scratch with a fine-tuned BERT model, finding the former to outperform the latter in predictive accuracy. The study also employed SHAP to enhance model interpretability, effectively providing both global and local explanations.

2.5 RESEARCH GAPS

Despite significant technological progress, the current body of research reveals several persistent gaps:

- **Limited Adoption of AutoML Platforms:** The integration of automated machine learning platforms, such as H2O.ai, remains relatively unexplored. This hampers the scalability and efficiency of experimentation and model deployment.
- **Insufficient Comparative Analysis:** There is a noticeable lack of comprehensive evaluations comparing traditional ML models with deep learning approaches using consistent datasets and performance metrics.
- **Neglect of Deployment and Practical Implementation:** While much of the literature emphasizes model development, few studies address the real-world deployment of predictive systems, particularly with respect to API integration and user accessibility.

2.6 THE LOAN LENS PROJECT: ADDRESSING THE GAPS

The “Loan Lens” project aims to address the aforementioned gaps through the following contributions:

- **Integration of H2O.ai AutoML** to streamline model training, tuning, and evaluation processes.

- **Comprehensive Model Comparison**, encompassing both ensemble-based ML algorithms and state-of-the-art DL frameworks to analyze performance differentials.
- **Deployment via FastAPI**, facilitating real-time prediction capabilities and enabling practical user interaction.
- **Incorporation of SHAP for Explainability**, ensuring transparency and interpretability in financial decision-making processes.

CHAPTER 3

PROPOSED METHODOLOGY

The proposed methodology for this project focuses on designing and implementing a robust machine learning pipeline for loan approval prediction, followed by deployment as a scalable API system. The methodology is divided into multiple stages, including data preprocessing, feature engineering, model development, evaluation, explainability, and deployment.

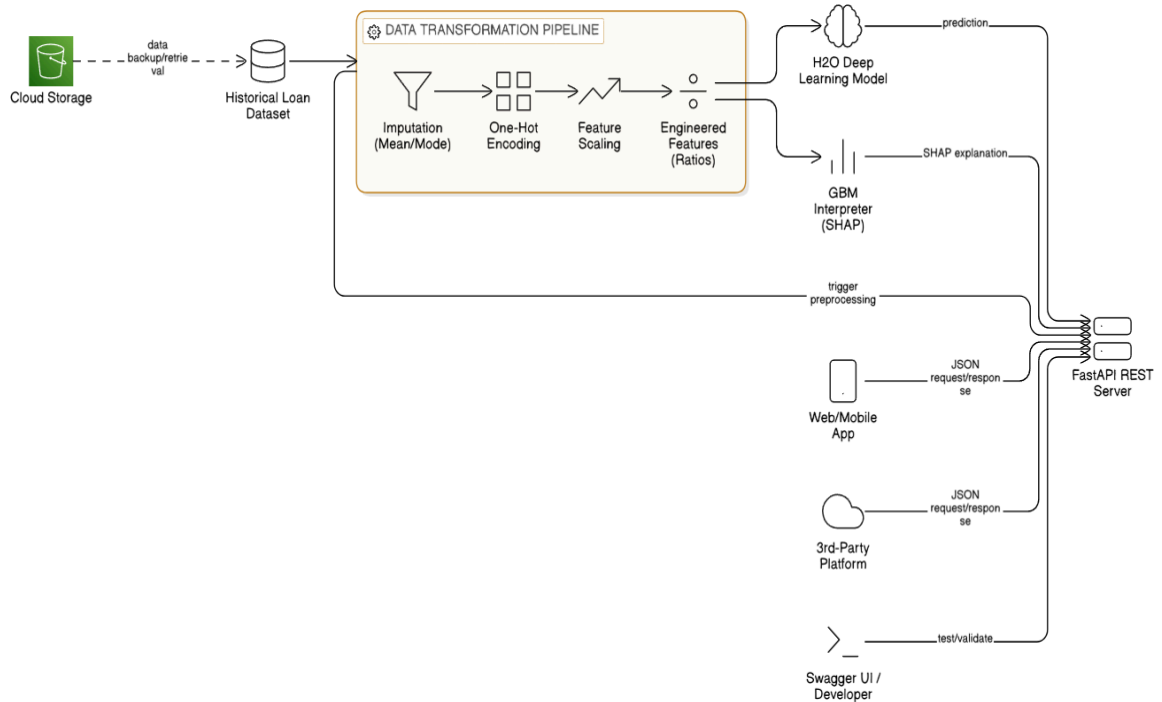


Figure 3.1: Architectural Framework for the Loan Lens Prediction Pipeline

3.1 DATA COLLECTION AND UNDERSTANDING

The first step involves obtaining a structured dataset containing historical loan application records. The dataset includes multiple attributes such as applicant income, credit score, employment history, loan amount, debt, credit history, and other financial indicators. Each record is associated with a target variable indicating whether the loan was approved or refused.

An initial EDA is performed to understand the distribution of features, identify correlations, detect outliers, and analyze class imbalance. This step helps in understanding which features significantly influence loan approval decisions.

3.2 DATA PREPROCESSING

Raw data often contains inconsistencies such as missing values, categorical variables, and skewed distributions. The preprocessing stage includes:

- **Handling Missing Values:** Missing numerical values are imputed using statistical measures such as mean or median, while categorical values are handled using mode or a separate category.
- **Categorical Encoding:** Categorical features such as loan purpose, home ownership, and employment status are transformed into numerical representations using techniques like one-hot encoding.
- **Feature Scaling:** Numerical features such as income, loan amount, and credit balance are normalized or standardized to ensure uniform contribution during model training.
- **Outlier Treatment:** Extreme values are detected and handled to prevent model bias.

This step ensures that the dataset is clean, consistent, and suitable for machine learning algorithms.

3.3 FEATURE ENGINEERING

Feature engineering is performed to enhance the predictive power of the model. New features may be derived from existing ones, such as:

- Debt-to-income ratio
- Credit utilization ratio
- Loan-to-income ratio

These engineered features provide better insights into the applicant's financial behavior and improve model performance.

3.4 MODEL TRAINING AND SELECTION

The project adopts a multi-model strategy using H2O.ai and Scikit-learn. Three main models are developed:

- **Gradient Boosting Machine (GBM):** Trained using H2O, this model is used primarily for interpretability and feature contribution analysis.
- **Deep Learning Model:** Also trained using H2O, this model serves as the main prediction engine due to its ability to capture complex non-linear relationships in the data.

- **XGBoost Model:** Implemented using Scikit-learn, this model is used for performance comparison and validation of results.

The dataset is split into training and testing sets, and models are trained using optimized hyperparameters. The goal is to achieve a balance between accuracy and generalization.

3.5 MODEL EVALUATION

The trained models are evaluated using performance metrics such as accuracy, precision, recall, and F1-score. Since loan prediction is a classification problem with risk considerations, different threshold strategies are applied to convert probability outputs into final decisions.

The deep learning model is selected as the final prediction model based on its superior performance. Threshold tuning is performed to control the trade-off between false approvals and false rejections, depending on the chosen evaluation metric.

3.6 EXPLAINABILITY MECHANISM

To ensure interpretability, SHAP is used to analyze feature contributions. Instead of applying explainability directly to the deep learning model, the GBM model is used for generating SHAP values due to its more stable and interpretable nature.

The SHAP analysis provides both global and local explanations, showing how each feature influences the prediction. This allows the system to justify decisions such as loan approval or rejection by highlighting key contributing factors like credit score, income, or debt

3.7 API Development and Deployment

The trained model is deployed using FastAPI, which provides a high-performance interface for real-time predictions.

- A REST API endpoint is created to accept input data in JSON format.
- The API processes the input, applies preprocessing steps, and feeds it into the trained model.
- The output includes both the prediction and feature-level explanations.

This architecture ensures that the system can be easily integrated with external applications such as web interfaces or financial platforms.

CHAPTER 4

RESULTS AND DISCUSSION

As we finish the data preprocessing and imputation, we will train the model. We will use 3 algorithms to compare the result, and use the top 2 as baseline models. The algorithms are gradient boosting, deep learning, and XGBoost. For the first 2 algorithms, I will use the H2O library. For the XGBoost algorithm, I will use the XGBoost library.

4.1 GBM MODEL

Model Metrics Binomial: gbm

Reported on test data.

MSE: 0.18219427094123514

RMSE: 0.4268422084813487

LogLoss: 0.5436201619563155

Mean Per-Class Error: 0.36949721360055027

AUC: 0.6918287015983052

AUCPR: 0.4791954127471261

Gini: 0.38365740319661046

Confusion Matrix (Act/Pred) for max f1 @ threshold = 0.24921493469792444

Table 4.1: Confusion Matrix for GBM

	loan_given	loan_refused	Error	Rate
loan_given	6919	5860	0.4586	(5860.0/12779.0)
loan_refused	1410	3618	0.2804	(1410.0/5028.0)

total	8329	9478	0.4083	(7270.0/17807)
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- The AUC score for this model is 0.6918. Based on the confusion matrix, it can be noted that the model is good in predicting a refused loan when the loan is supposed to be refused. However, the performance is not too good for predicting loans which are given. The number of false positives are almost the same as the number of true positives.
- Before calculating the metrics, the Shap summary plot is generated to see what features are the most important in affecting the result of the model's prediction.

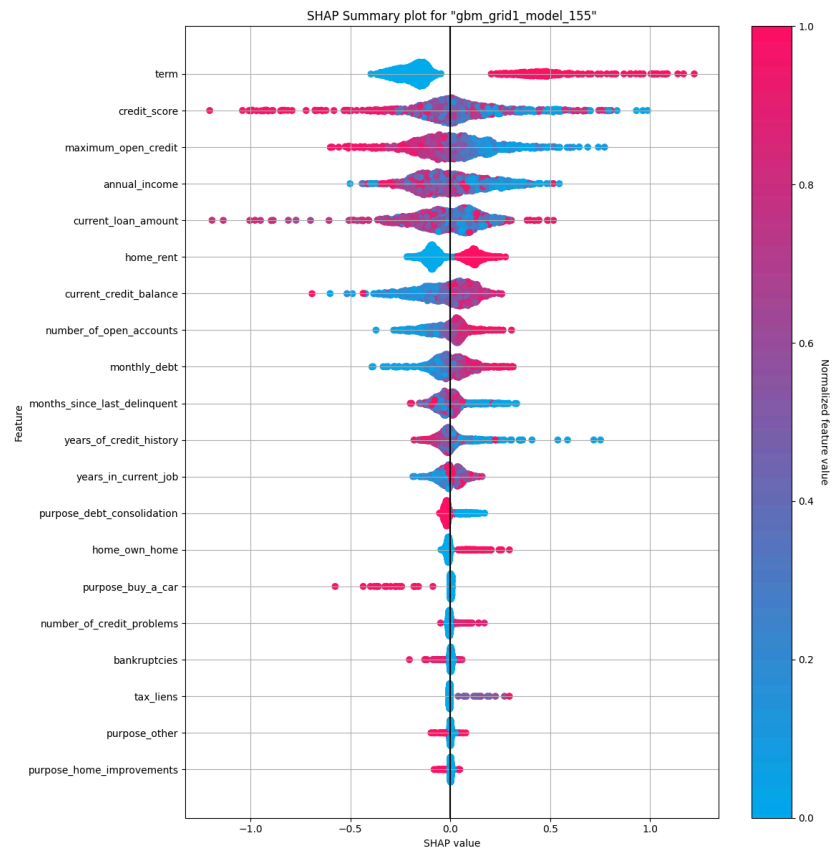


Figure 4.1: SHAP Summary Plot for the Gradient Boosting Machine (GBM) Model

SHAP summary can be interpreted as follows:

1. The most influential features are: **term**, **credit_score**, **maximum_open_credit**.
2. These three features contribute the most to the variability in the predicted outcome and are crucial for model interpretability and optimization.
3. Features such as **purpose_debt_consolidation**, **purpose_home_improvements**, and **purpose_other** show minimal SHAP value distribution, indicating that their contribution to the prediction is negligible.
4. In SHAP plots, the **X-axis** indicates the SHAP value, which reflects the impact of each feature on the prediction:
 - Values > 0 push the prediction towards **loan approval** (Class 1).
 - Values < 0 push the prediction towards **loan rejection** (Class 0).
 - The **color gradient** represents the feature value: red denotes high values and blue denotes low values.
5. A contradiction is noted when interpreting the feature **term**:
 - The SHAP summary suggests that **longer loan terms (red dots)** tend to increase the likelihood of loan approval.
 - However, based on external validation (e.g., class distribution or bar charts), **shorter-term loans** are generally more likely to be approved.
 - This discrepancy suggests the model might have learned an unexpected pattern, which warrants further cross-analysis between the SHAP summary plot and actual class distribution statistics.

6. The **beeswarm nature** of the plot shows both feature importance and interaction. A dense spread across the SHAP value axis for key features suggests high variability and contextual dependence.

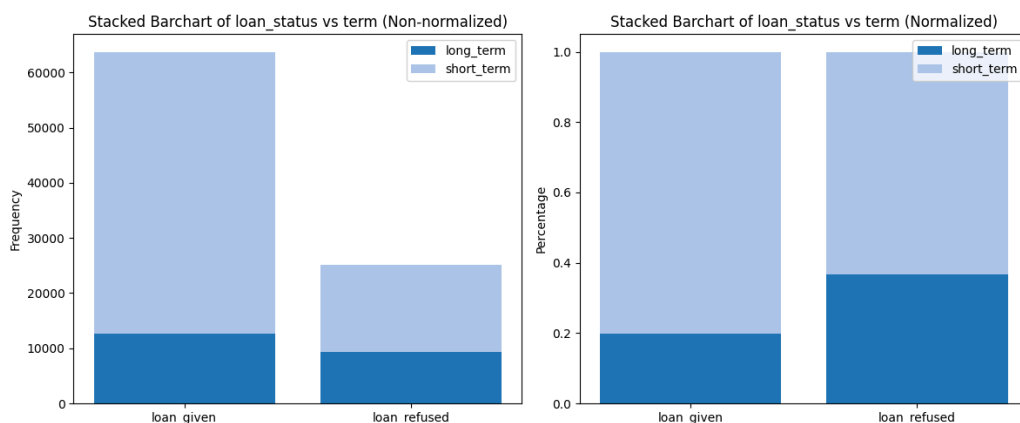


Figure 4.2: Stacked Bar Chart Analysis of Loan Status Relative to Term Duration (Non-normalized and Normalized)

7. Model transparency and fairness can be improved by using this SHAP summary, especially to justify model behavior to stakeholders.

MODEL SUMMARY:

The F1 score for GBM is 0.7368699813152678

The precision score for GBM is 0.9829205973919817

The recall score for GBM is 0.7272684420894501

The AUC score for GBM is 0.6918287015983052

The key takeaways from the model summary are:

1. The F1 score indicates that the model can achieve a good balance between making correct positive predictions and capturing positive instances.
2. Based on the precision score, this model is able to predict 98.29% approved loans

when the loans are supposed to be approved. This is a good score for the model.

3. Based on the recall score, this model can recognize 72.72% of all eligible loan applicants. Although this number is considerably good, this means that the business has the chance of losing 27.28% eligible loan applicants.
4. The AUC score indicates that the model's power to discriminate between positive and negative cases (loan given and loan refused) is somewhat good.

4.2 DEEP LEARNING MODEL

Model Metrics Binomial: deeplearning

Reported on test data.

MSE: 0.18485588386780527

RMSE: 0.42994869911165595

LogLoss: 0.551741004739872

Mean Per-Class Error: 0.374275222382485

AUC: 0.6780014935377457

AUCPR: 0.46524590064364296

Gini: 0.3560029870754915

Confusion Matrix (Act/Pred) for max f1 @ threshold = 0.27419587432435266

Table 4.2: Confusion Matrix for DL Model

	loan_given	loan_refused	Error	Rate
loan_given	7018	5761	0.4508	(5761/12779)
loan_refused	1497	3531	0.2977	(1497/5028)
total	8515	9292	0.4076	(7258/17807)

- The AUC score of the deep learning model is lower compared to the gradient boosting model. So, we have to plot the feature importance. Since SHAP is not supported here, we will go straight to plot the bar chart of the feature importance.

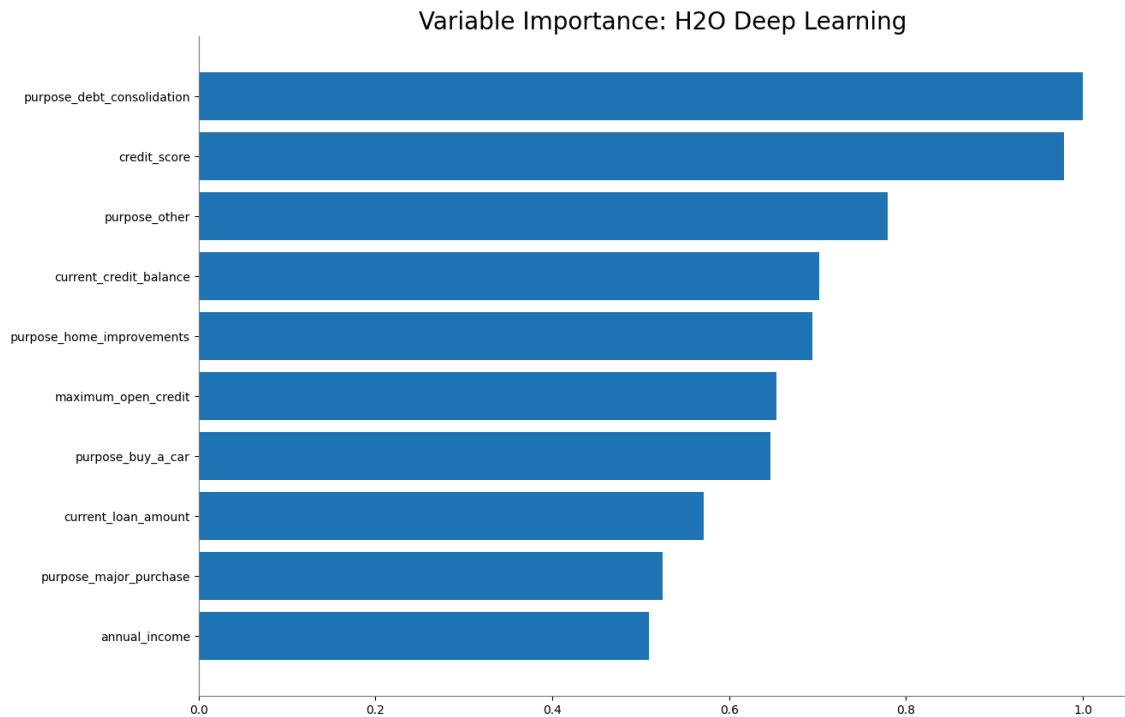


Figure 4.3: Variable Importance Plot for the H2O Deep Learning Model

- The result of this feature importance plot is a bit different compared to the gradient boosting one. The most important feature is **purpose_debt_consolidation**, followed by **credit_score** and **annual_income**.

MODEL SUMMARY:

The F1 score for GBM is 0.7368699813152678

The precision score for GBM is 0.9829205973919817

The recall score for GBM is 0.7272684420894501

The AUC score for GBM is 0.6918287015983052

Model Summary is as follows:

1. The F1 score indicates that the model can achieve a good balance between making correct positive predictions and capturing positive instances.
2. Based on the precision score, this model is able to predict 98.29% approved loans when the loans are supposed to be approved. This is a good score for the model.
3. Based on the recall score, this model can recognize 72.72% of all eligible loan applicants. Although this number is considerably good, this means that the business has the chance of losing 27.28% eligible loan applicants.
4. The AUC score indicates that the model's power to discriminate between positive and negative cases (loan given and loan refused) is somewhat good.

4.3 XGBOOST MODEL

MSE: 0.2640

RMSE: 0.5138

Mean Per-Class Error: 0.4373

AUC: 0.5627

AUCPR: 0.465

Gini: 0.1254

Table 4.3: Confusion Matrix for XGBoost Model

	loan_given	loan_refused	Error	Rate
loan_given	12272	460	0.0361	(460/12732)
loan_refused	4234	816	0.8384	(4234/5050)

total	16506	1276	0.2640	(4694/17782)
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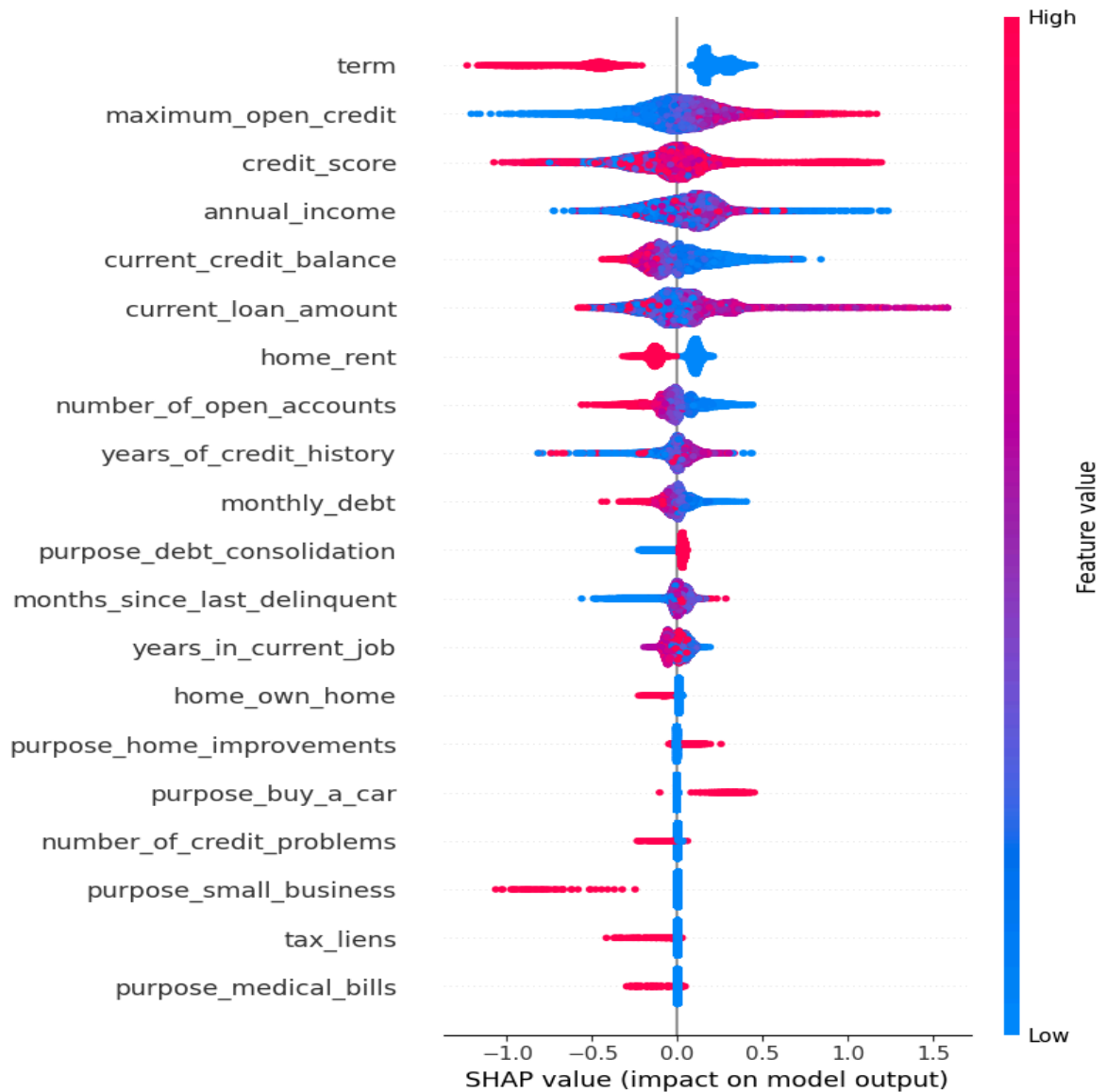


Figure 4.4: SHAP Summary Plot for the Extreme Gradient Boosting (XGBoost) Model

The key pointed of the SHAP Plot is as:

1. The top 3 features of this model are **term**, **maximum_open_credit**, and **credit score**.

2. Beginning on the **purpose_debt_consolidation** to **purpose_medical_bills**, these features don't affect the prediction much, since the examples are mostly located around 0.
3. When the loans are long term, it makes the model predict the application as not given (loan_refused). Whereas when the loans are short term, it makes the model predict the application as given (loan_given).
4. Higher maximum open credit is also associated with the model that predicts the loan application to be given.
5. A higher credit score is also associated with the model that predicts the loan application to be given, although there are obvious examples that people with higher credit scores also get their loans rejected as well.

The SHAP summary plot is similar with the gradient boosting one, especially for the top important features. But assuming that the SHAP summary plot built-in the H2O library is indeed inverted, then it can be said that it's very similar to the gradient boosting one.

MODEL SUMMARY:

The F1 score for XGBoost is 0.6743201692727949

The precision score for XGBoost is 0.7139549166882692

The recall score for XGBoost is 0.7360251940164211

The AUC score for XGBoost is 0.5627273603891964

The key takeaways from the model summary are:

1. The F1 score indicates that the model can achieve a somewhat good balance between making correct positive predictions and capturing positive instances.
2. Based on the precision score, this model is able to predict 71.39% approved loans when the loans are supposed to be approved. This is a good score for the model.

3. Based on the recall score, this model can recognize 73.6% of all eligible loan applicants. Although this number is considerably good, this means that the business has the chance of losing 27.28% eligible loan applicants.
4. The AUC score indicates that the model's power to discriminate between positive and negative cases (loan given and loan refused) is not too good. It's too close to 0.5 which indicates that the model is closer to a random guess than an accurate prediction.

4.4 COMPARISON BETWEEN 3 MODELS:

We have to compare all the above mentioned 3 models based on F1 Score, Precision, Recall, AUC Score. It will help to choose the best model for our Loan Prediction Tasks.

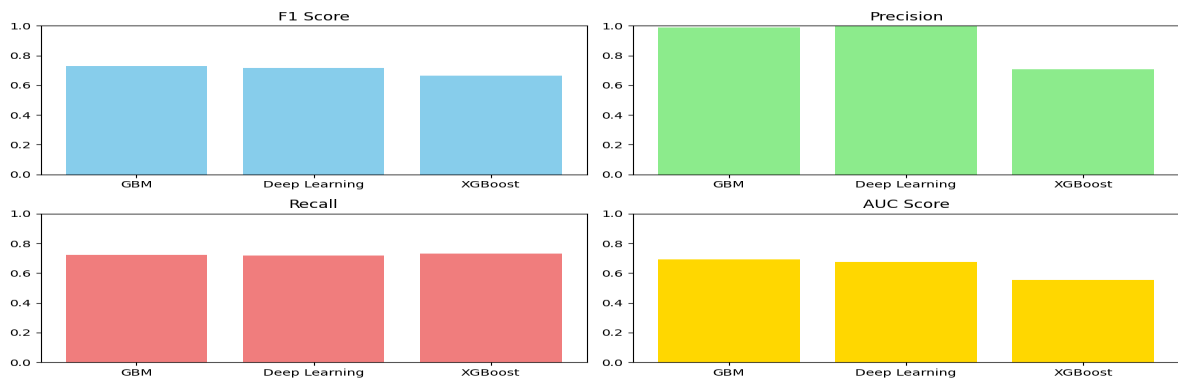


Figure 4.5: Comparative Analysis of Performance Metrics (F1, Precision, Recall, and AUC)

The resulting base models and models generated from the experiments are a bit mixed. However, it's obvious that the models from the experiment with both original and engineered features performed worse than the base model and the one with no multicollinearity.

Table 4.4: Comparative Performance Summary of All 3 Models

	Model	F1 Score	Precision	Recall	AUC Score
0	GBM	0.728471	0.987460	0.722694	0.690368
1	Deep Learning	0.718002	0.995540	0.717329	0.676931
2	XGBoost	0.665310	0.709298	0.733465	0.552384

The best AUC scores are generated from the base models, with the highest being the base GBM model. However, the GBM model with no multicollinearity holds the best precision score. In terms of scores, this model outperforms the best GBM model apart from the AUC score. The deep learning model with no multicollinearity has the best F1 and recall scores. However, although the precision score is quite high, it dropped from the base deep learning model.

Therefore, the best model needed to choose depends on the requirements of the user. If the user demands a high precision score, GBM model with no multicollinearity can be chosen. If the user wants the highest recall or F1 score, then the deep learning model with no multicollinearity. If the concern is the AUC score, the goto will be the base GBM model.

For this project, the chosen one is the deep learning model with no multicollinearity (dl_grid_model_66) since in general, it performs quite well. Although the precision and AUC scores have to be sacrificed.

The model explainability are as follows:

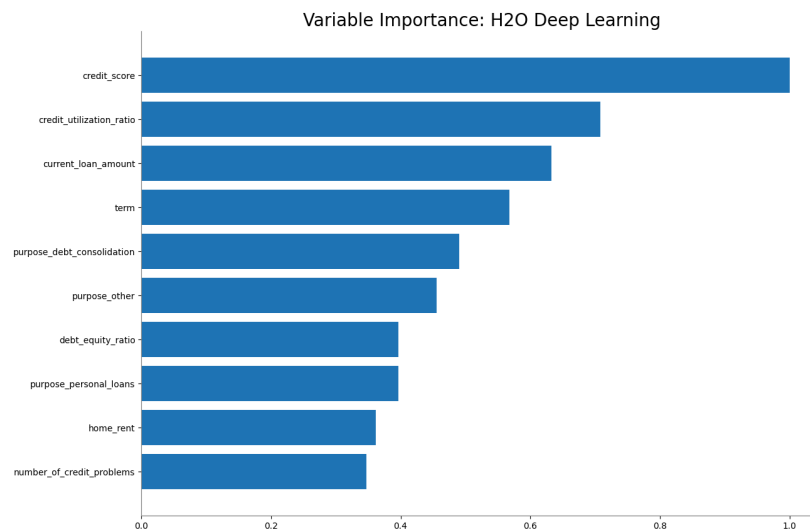


Figure 4.6: Variable Importance Plot for the Optimized Deep Learning Model

The barchart indicates that the top 3 most important features are **credit_score**, **credit_utilization_ratio**, and **current_loan_amount**. They affect the model's prediction the most. It has to be noted, however, that deep learning models are considerably more challenging to interpret. For this model, as far as it is known, the H2O varimp plot works better with a tree-based model compared to a deep learning model. To interpret this model well, we might need to create it from scratch using tensorflow or pytorch, and use SHAP vales or Grad-cam techniques to do so.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

This project presented the design and implementation of a machine learning-based loan classification system aimed at improving the efficiency and consistency of loan approval decisions. The system utilized a structured dataset containing financial and credit-related attributes to train and evaluate multiple predictive models. A comparative analysis was conducted using Gradient Boosting Machine (GBM), XGBoost, and a deep learning model. Among these, the deep learning model demonstrated superior performance in capturing complex, non-linear relationships within the data and was therefore selected as the final prediction model.

To address the challenge of interpretability commonly associated with deep learning models, an explainability mechanism was incorporated using SHAP (SHapley Additive exPlanations). Instead of applying SHAP directly to the deep learning model, the GBM model was used to generate feature contribution values due to its more stable and interpretable nature. This hybrid approach enabled the system to maintain high predictive accuracy while also providing meaningful insights into how individual features such as credit score, income, and debt influence the final decision.

The system was further deployed as a RESTful API using FastAPI, allowing real-time predictions based on user-provided JSON input. The integration of preprocessing, prediction, threshold-based decision-making, and explainability within a single pipeline demonstrates the practical applicability of the solution. The project highlights how machine learning can support data-driven decision-making in financial systems, reducing manual effort and improving consistency. However, it also emphasizes the importance of threshold selection and interpretability when deploying such models in real-world scenarios.

5.2 FUTURE SCOPE

While the current system demonstrates strong performance and practical applicability, several improvements can be explored in future work to enhance its robustness and real-world relevance. One major area of improvement is the use of real-world financial datasets instead of synthetically generated data. Real-world data would introduce more complex patterns and variability, enabling the model to generalize better and perform more reliably in practical scenarios.

Another important extension is the implementation of dynamic thresholding strategies. Instead of relying on fixed metrics such as precision or F1-score, adaptive thresholds can be designed based on risk levels, allowing the system to categorize applications into approval, rejection, and manual review zones. This would better reflect real-world lending practices followed by financial institutions.

The explainability component can also be further enhanced by integrating direct SHAP support for deep learning models or by incorporating additional interpretability techniques such as LIME. This would provide more accurate, model-specific explanations rather than relying on surrogate models.

From a system perspective, the project can be extended by developing a complete user interface and deploying the application on cloud platforms for broader accessibility. Integration with databases can enable storage of historical predictions and support continuous learning. Additionally, implementing monitoring mechanisms for data drift and model performance degradation would ensure long-term reliability and maintain model accuracy over time.

Overall, the project provides a strong foundation for building intelligent financial decision systems, and with further enhancements, it can evolve into a more scalable, reliable, and industry-ready solution.

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APPENDIX 1

A. API DESCRIPTION

The suggested system is implemented as a RESTful API using FastAPI. This API allows users to provide loan application information in JSON format and get the prediction along with the explanation of features.

The API takes the input data, preprocesses it, makes the inference with the model, and returns the results as structured JSON.

There are two principal endpoints in the system:

- /predict – Gives the loan acceptance result
- /predict_explanation – Provides both the prediction and the feature contribution

B. SAMPLE REQUEST AND RESPONSE

Sample Input (Request):

```
{  
  
  "data": {  
  
    "current_loan_amount": 8000,  
  
    "term": "Short Term",  
  
    "credit_score": 745,  
  
    "years_in_current_job": 8,  
  
    "home_ownership": "Own Home",  
  
    "annual_income": 145000,
```

```

    "purpose": "Home Improvements",
    "monthly_debt": 520.33,
    "years_of_credit_history": 18.2,
    "months_since_last_delinquent": 0,
    "number_of_open_accounts": 12,
    "number_of_credit_problems": 0,
    "current_credit_balance": 62000,
    "maximum_open_credit": 125000,
    "bankruptcies": 0,
    "tax_liens": 0
  },
  "threshold": {
    "threshold_metrics": "f1"
  }
}

```

Sample Output (Response):

```

{
  "prediction": {
    "prediction": ["Loan Approved"]
  },
  "feature_influence": {

```

```

"credit_score": 0.90,

"term": 0.48,

"annual_income": 0.03,

"monthly_debt": -0.01

}

}

```

C. FEATURE DESCRIPTION

FEATURE	DESCRIPTION
Credit Score	Numerical representation of applicant's creditworthiness
Annual Income	Total yearly income of applicant
Monthly Debt	Monthly financial obligations
Loan Amount	Requested loan amount
Years of Credit History	Duration of credit usage
Home Ownership	Type of residence (Rent/Own)
Purpose	Reason for taking the loan

D. SYSTEM REQUIREMENTS

- Python 3.8 or above
- Libraries: H2O, Pandas, NumPy, Scikit-learn
- Framework: FastAPI
- Operating System: Windows/Linux

E. NOTE

The appendix provides implementation-level details, including API structure and sample inputs/outputs, to ensure clarity, reproducibility, and understanding of the deployed system.

Enhancing Industrial Internet of Things Security: Challenges and Solutions for AI-Driven Intrusion Detection Systems

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Abstract— The Industrial Internet of Things (IIoT) is a great factor for the enhancement of the entire supply chain and industrial operations through the implementation of smart devices, big data analytics, and artificial intelligence. However, the developments made in these areas also bring about challenges in cybersecurity. The issue is that connected devices, in general, are the source of increased cyber risks. Among the measures that can be taken to deal with the negative impacts of these security risks, an essential one is the use of Intrusion Detection Systems (IDS), which help to greatly mitigate and manage the risks by constantly monitoring network traffic and system activity in order to quickly recognize and counteract IIoT cyber threats and attacks. The present investigation deals with the AI-based IDSs that provide security for IIoT. In addition, the differences in IIoT, the need for quick response, the lack of resources, and the importance of IIoT assets are among the hurdles that researchers must cope with. These challenges are still unsolved, and the paper suggests enhancements in the efficiency and dependability of IIoT IDS as one of the research areas. Besides, the paper discusses the role of machine learning and artificial intelligence in improving detection accuracy and turning back new cyber threats. By presenting difficulties and solutions, the paper provides insights into the capability of AI-based intrusion detection systems to recognize, react to, and lessen sophisticated cyber-attacks and threats targeting IIoT-controlled critical infrastructure systems.

Keywords— IIoT, artificial intelligence, machine learning, IDS, deep learning

I. INTRODUCTION

The Internet of Things (IoT) is revolutionizing our interaction with technology by providing a direct internet connection to everyday objects. Health and wellness, agriculture, manufacturing, and urban sustainability are just a few of the domains where IoT applications are already having a positive impact. They not only stimulate innovations but also facilitate the streamline of processes. Industrial Internet of Things (IIoT) is a sector of the IoT intended for industrial use that unlocks the modern digital world by joining industrial

machines with intelligent sensors, high-volume data processing, and machine learning, ultimately leading to the optimization of production and supply chain processes [1]. The IIoT ecosystem can be utilized for various purposes including smart manufacturing [5], predictive maintenance [2], asset monitoring and management [3], and energy management [4]. However, the large number of IIoT devices raises the stakes in terms of security. Cybersecurity risks such as data leaks, ransomware, and DoS attacks are the uninvited guests that can severely undermine the very foundation of IIoT infrastructure. Cybersecurity Ventures forecasts that cybercrime will impose a global financial burden of \$10.5 trillion per year by 2025, which only proves how urgent it is to have strong security measures in place [6]. A survey conducted in 2023 also reported that 70% of the companies in the industrial sector had experienced a security breach during the previous year [7]. To cope with these security challenges, Intrusion Detection Systems (IDS) are increasingly being adopted and recognized as a significant security measure. Instead of looking for Wi-Fi hackers, IDS focuses on detecting breaches of network policy, e.g. monitoring traffic and system activities for any signs of unauthorized and suspicious behaviors. They become an integral part of identifying and handling potential security issues while ensuring that the information systems are protected against both unauthorized access and attacks [8]. In IIoT management specifically, the role of IDS is clarified due to the critical nature of industrial systems and the high risk associated with their security gaps.

To effectively deal with these problems, IDS tailored for IIoT settings are indispensable. The advanced IDS for IIoT uses machine learning and artificial intelligence to raise the detection accuracy and bring down the number of false positives, thus, very cleverly adapting to the always changing threat environment [9]. Studies show that the installation of intrusion detection systems in Industrial Internet of Things environments alone can lead to a reduction in the number of cyber-attacks by as much as 60%. Moreover, the surveillance and learning aspects of the IDS ensure that the industrial systems are always protected against new and developing

threats. Continuing research and development in IDS technologies are necessary for keeping up with the changing landscape of cyber threats. The objective of this research is to analyze the current issues, innovations, and the AI-driven IDS solutions in the IIoT. It also provides a detailed review of the ways AI methods enhance IIoT security, highlights the restrictions, and recommends future research and implementation. The next parts of this survey are structured as follows: Section 2 investigates the configuration of the Industrial Internet of Things (IIoT). Section 3 gives a recap of Intrusion Detection Systems. Section 4 centers around the current work on AI-driven IoT IDS. Section 5 outlines some of the remaining major research challenges, while section 6 wraps up the study and outlines possible paths for future exploration.

II. BACKGROUND

A. The IIoT System Architecture

The IIoT can be defined as the networked integration of devices, sensors, and industrial systems. This technology has the potential to gather and analyze data which would consequently benefit the industrial sector through increased efficiency. The IIoT can be considered a form of IoT that incorporates cloud and front-end computing to enhance automation. Security in IIoT has recently become a subject of much interest, reminiscent of the situation surrounding IoT [10]. The objectives of IIoT security solutions include protecting devices and ensuring secure data transmission by making use of new programs and techniques. IIoT is a blessing for the industries as it allows for the merging of the physical and virtual worlds which leads to the enhancement of the operational efficiencies [11]. The Industrial Internet of Things (IIoT) has various protocols available for connecting to the Internet. The IIoT architecture is shown in Figure 1.

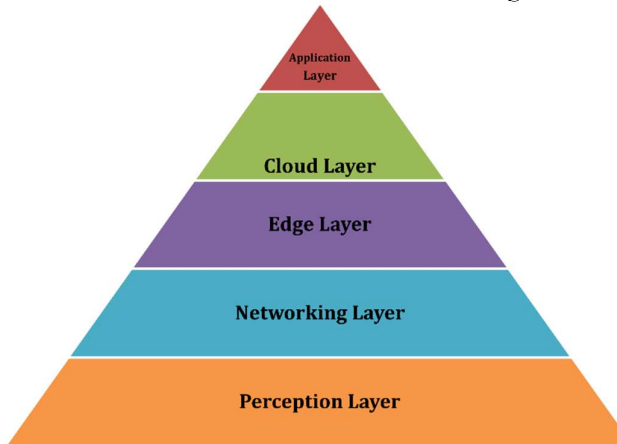


Fig. 1 Architecture of IIoT system

It consists of five layers: perception, networking, Edge or Fog Layer, cloud and application. The tangible components are addressed in the initial stratum [12]. The second layer encompasses communication protocols that enable the transfer of data to the third layer, which is optional but necessary, does calculations near the devices to cut down on latency. This layer is used for pre-processing data (filtering, compression, and finding anomalies) to lower the strain on the cloud and speed up response times. The fourth layer

provides comprehensive analytics, data storage, and scalability. The application layer is the top layer, where processed data is utilized to make industrial applications. k feature. All tiers of the IIoT are vulnerable to attacks and intruders. The major cybersecurity threats comprised DoS, DDoS as well as deception and spamming, while MiTM assaults were carried out sporadically. Many corporations make use of IDSs in order to pinpoint malevolent attacks and provide protection to the nodes and networks of IIoT [13].

Intrusion Detection Systems

Data integrity, confidentiality, and security in networks are greatly influenced by IDS. Therefore, it plays a major role in IIoT security. The main functions of an IDS include not only the very important detection but also preventing, responding to, and reporting the harmful acts in IIoT networks. It is possible to categorize and place the IDSs at any level of system architecture. Network monitoring employs packet data collection [14]. Cybercriminals often resort to using either malware or network packet analysis to retrieve sensitive information. Attacks could be directed at either the server that processes all network transactions or the workstation that is engaging in network activity. There might be opportunities for the exploitation of system vulnerabilities. Indeed, machine learning together with deep learning form the basis for the advanced intrusion detection systems that can very accurately identify network threats [15]. Data for intrusion analysis can come from multiple sources. Depending on the information sources, IDSs can be either host-based or network-based. The identification of intrusive and malicious activities will be through the examination of the infrastructure-specific technique in the case of the IDS [16]. In Figure 2, the analytical method indicates that the IDSs fall into the categories of anomaly-based and misuse-based.

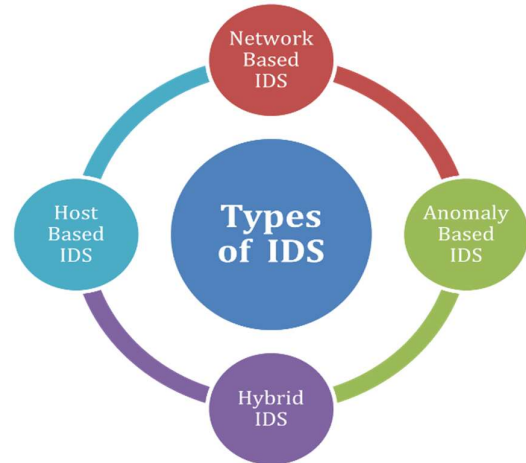


Fig. 2. IDS taxonomy system

III. RELATED WORK

Many frameworks and models have been proposed for identifying IIoT categories at the levels of device, network, and database. The next part of the paper is about discussing the different solutions where machine learning and deep learning techniques have been used.

O. Almomani et al. [17] An IDS framework has been implemented with the different classifiers. The feature selection process involved three algorithms such as Genetic Algorithm (GA) using UNSW-NB15 dataset. The performance of GA-J48, GWO-J48, and FFA-J48 was measured in terms of accuracy where they got 86.874%, 85.676%, and 86.037% respectively. Likewise, accuracy ratings of 86.387%, 84.485%, and 85.429% were recorded for GA-SVM, GWO-SVM, and FFA-SVM, respectively. Therefore, the combination of J48 and SVM algorithms rendered remarkable results.

Z. Chen et al. [18] utilized different XGBoost thresholds as a method for feature selection. The proposed model was tailored to feature selection and loss function thresholds that would be the best for our model. The MIX_LSTM model outperforms DL, ML, and even the finest algorithms at detecting rare Industrial Internet of Things risks.

A. Yazdinejad et al. [19] deep learning model based on Long Short-Term Memory (LSTM) and Auto-Encoder (AE) in ensemble is applied to detect abnormal behavior for cyber threat hunting in IIoT. The model employs unbalanced IIoT datasets from Gas Pipeline (GP) and Secure Water Treatment (SWaT). Traditional machine learning classifiers such as Random Forest give lower accuracy than proposed model.

S. Tharewal et al. [20] introduced a deep reinforcement learning method was presented which could be regarded as a very efficient near-end strategy optimization technique for the IDS. The DRL-IDS system uses LightGBM to quickly and effectively extract the most pertinent feature set from the IoT data and deploys deep learning techniques for carrying out the intrusion detection process. The proposed solution has been able to detect 99 percent of the IoT network attacks using some publicly available dataset. Moreover, it has a false acceptance rate of 0.9%.

T. Gaber et al. [21] introduced a new approach that featured the use of PSO and BA for selecting the characteristics, while Random Forest (RF) was the classifier used for IIoT network traffic classification. The model was tested on the dataset WUSTL-IIOT-2021. The objective was to select the best one by comparing the results of the two feature selection methods. RF+BA was the classifier that performed the best.

H. V. Krishna and B.Samatha [22] proposed method that employs ensemble classifiers for the purpose of detecting attacks. IIoT networks are constantly changing and interconnected. Hence, the proposed system is capable to cope with them. The enhanced Adaboost algorithm works on the principle of weighted error and by giving priority to misclassified occurrences, it not only increases the accuracy of the RF classifier but also gives rise to the formation of diverse and flexible clusters. The experimental results confirm that it is capable of rapidly detecting and preventing intrusions.

Z.Huma et al. [23] introduced the utilization of HDRaNNs for pinpointing IIoT network breaches has been researched extensively. The architecture is composed of deep random neural networks and multilayer perceptrons that apply dropout regularization. The HDRaNN model classified the DS2OS

and UNSW-NB15 datasets with respective accuracies of 98% and 99% after having identified 16 types of cyberattacks.

IV. CHALLENGES FACED BY IDS IN THE INDUSTRIAL INTERNET OF THINGS

The links of IIoT enable machine-to-machine communication in diverse fields [23]. Industrial IoT applications are very important but are sometimes unreliable due to the differences in technology, constantly changing environments, and the large amount of data generated. IIoT has its advantages as well as disadvantages. The ability to quickly analyze large volumes of data from Industrial Internet of Things (IIoT) devices requires many critical features. Examples are proper device maintenance, network security, data confidentiality, access control, machine and human error. Not tackling these problems may have a negative impact on the performance of the industry.

The IIoT brings along a new set of operational, environmental, and architectural limitations that make the Intrusion Detection System design, deployment, and use very difficult. The main IDS issues are shown in Figure 3. A list of the main problems is presented below [24][25]:

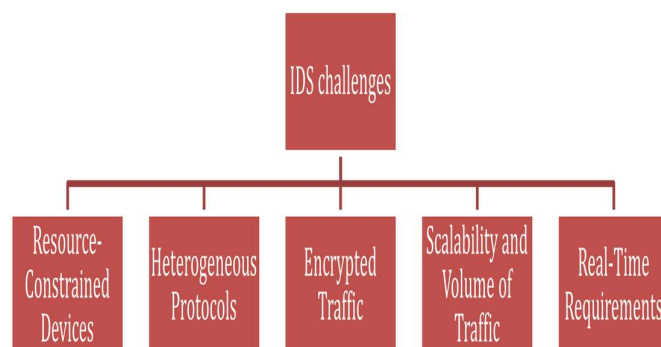


Fig. 3. IDS challenges

- **Devices with limited resources:** Many IIoT endpoints, such as sensors, actuators, PLCs, and RTUs, don't have a lot of processing power, memory, battery life, or storage. Traditional IDS that use signatures or deep packet inspection need a lot of processing power and memory, thus they can't be installed directly on edge devices. We need lightweight models for detecting anomalies, but getting good accuracy with less resources is still hard.
- **Different Protocols and Old Systems:** IIoT settings employ both old and new industrial protocols. Some examples of new protocols include MQTT, OPC UA, CoAP, and DDS. Some examples of old protocols are Modbus/TCP, PROFINET, EtherNet/IP, S7comm, DNP3, and IEC 61850. Many old protocols don't have encryption, authentication, or adequate checks for message integrity built in. Keeping up-to-date

signature databases for dozens of proprietary and poorly described protocols is quite hard.

- Requirements for Real-Time and Deterministic: Industrial control systems need communication that is both low-latency and predictable, usually in the millisecond range. Deep packet inspection or complicated behavioral analysis might cause delays or jitters that are not acceptable. IDS must work in a way that is completely non-intrusive (passive) or with assured limited latency.
- How much traffic it can handle and how big it can become: IIoT networks may include tens or hundreds of thousands of devices that provide telemetry and control data all the time. Centralized IDS systems might become sites of failure and bottlenecks. We need distributed and hierarchical IDS designs, but they make things more complicated and take more time to administer.
- Traffic that is encrypted: More people are using TLS/DTLS, IPsec, and encryption that is particular to a protocol (like OPC UA with message security). This makes payload-based detection useless. Conventional deep packet inspection fails unless expensive man-in-the-middle decryption proxies are used, which are often deemed undesirable in operational technology settings owing to certification and safety concerns.

V. FUTURE DIRECTIONS FOR IIOT

The use of AI technology in IIoT ecosystem security through intrusion detection is still a long way to go, though it has been a gradual process. The challenges that need to be tackled further are crucial in developing secure, scalable and reliable IDS products for the advanced industrial systems [26][27].

- IDS with Edge Intelligence and Lightweight : The future exploration of this area should focus on the development of lightweight and computationally efficient AI models for IIoT edge devices. To achieve real-time intrusion detection with low latency, energy, and communication overhead, model pruning, quantization, and TinyML will be utilized.
- Federated, Private Learning: Federated learning along with decentralized AI frameworks will facilitate the overcoming of data privacy and bandwidth problems. These techniques will allow the remote IIoT nodes to train the IDS models without sharing the raw data, hence increasing the scalability while at the same time safeguarding the industrial data that is sensitive [28].
- Explainable, reliable AI : IIoT systems are critical to safety, thus the future IDS must incorporate XAI. The detection decision explanations that are transparent and comprehensible not only enhance the operator's confidence and incident response but also give industrial cybersecurity compliance a boost.
- Self-learning and adaptive detection mechanisms: Traditional IDS are not equipped to handle zero-day and evolving attacks. The future AI-driven IDS must apply continuous learning and self-adaptive techniques to make the detection models update according to new traffic patterns without losing stability or performance [29].
- Integrating Digital Twins :The combination of IDS and digital twin technologies results in proactive security. Digital twins can simulate cyberattacks, generate authentic

training data, and validate security measures before they are applied in the industrial systems [30].

VI. CONCLUSION

The swift expansion of the IIoT opens new avenues for innovation and higher efficiency in industrial operations like never before. Still, the larger area of possible attacks and the vital importance of industrial systems also create a lot of difficulties in terms of cybersecurity. The integration of AI in IDS enhances their efficacy and guarantees their capacity to address the specific challenges of IIoT, including device heterogeneity and real-time operational demands. In this survey, we analyzed ML and DL methods employed in the development of Intrusion Detection Systems (IDS) over the past five years, specifically from 2020 to 2026. Although progress has been achieved, numerous areas still require further research and development. Enhancing the robustness of AI-driven IDS against advanced assaults, improving their scalability for extensive IIoT deployments, and ensuring their reliability across diverse industrial environments are essential areas that warrant further investigation. By progressing these technologies and overcoming their existing limitations, we will be able to greatly enhance the security and reliability of industrial systems, which in turn will give a guarantee to the future safe and efficient operation of IIoT infrastructures.

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Enhancing Industrial Internet of Things Security: Challenges and Solutions for AI-Driven Intrusion Detection Systems

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Abstract— The Industrial Internet of Things (IIoT) is a great factor for the enhancement of the entire supply chain and industrial operations through the implementation of smart devices, big data analytics, and artificial intelligence. However, the developments made in these areas also bring about challenges in cybersecurity. The issue is that connected devices, in general, are the source of increased cyber risks. Among the measures that can be taken to deal with the negative impacts of these security risks, an essential one is the use of Intrusion Detection Systems (IDS), which help to greatly mitigate and manage the risks by constantly monitoring network traffic and system activity in order to quickly recognize and counteract IIoT cyber threats and attacks. The present investigation deals with the AI-based IDSs that provide security for IIoT. In addition, the differences in IIoT, the need for quick response, the lack of resources, and the importance of IIoT assets are among the hurdles that researchers must cope with. These challenges are still unsolved, and the paper suggests enhancements in the efficiency and dependability of IIoT IDS as one of the research areas. Besides, the paper discusses the role of machine learning and artificial intelligence in improving detection accuracy and turning back new cyber threats. By presenting difficulties and solutions, the paper provides insights into the capability of AI-based intrusion detection systems to recognize, react to, and lessen sophisticated cyber-attacks and threats targeting IIoT-controlled critical infrastructure systems.

Keywords— IIoT, artificial intelligence, machine learning, IDS, deep learning

I. INTRODUCTION

The Internet of Things (IoT) is revolutionizing our interaction with technology by providing a direct internet connection to everyday objects. Health and wellness, agriculture, manufacturing, and urban sustainability are just a few of the domains where IoT applications are already having a positive impact. They not only stimulate innovations but also facilitate the streamline of processes. Industrial Internet of

Things (IIoT) is a sector of the IoT intended for industrial use that unlocks the modern digital world by joining industrial machines with intelligent sensors, high-volume data processing, and machine learning, ultimately leading to the optimization of production and supply chain processes [1]. The IIoT ecosystem can be utilized for various purposes including smart manufacturing [5], predictive maintenance [2], asset monitoring and management [3], and energy management [4]. However, the large number of IIoT devices raises the stakes in terms of security. Cybersecurity risks such as data leaks, ransomware, and DoS attacks are the uninvited guests that can severely undermine the very foundation of IIoT infrastructure. Cybersecurity Ventures forecasts that cybercrime will impose a global financial burden of \$10.5 trillion per year by 2025, which only proves how urgent it is to have strong security measures in place [6]. A survey conducted in 2023 also reported that 70% of the companies in the industrial sector had experienced a security breach during the previous year [7]. To cope with these security challenges, Intrusion Detection Systems (IDS) are increasingly being adopted and recognized as a significant security measure. Instead of looking for Wi-Fi hackers, IDS focuses on detecting breaches of network policy, e.g. monitoring traffic and system activities for any signs of unauthorized and suspicious behaviors. They become an integral part of identifying and handling potential security issues while ensuring that the information systems are protected against both unauthorized access and attacks [8]. In IIoT management specifically, the role of IDS is clarified due to the critical nature of industrial systems and the high risk associated with their security gaps.

To effectively deal with these problems, IDS tailored for IIoT settings are indispensable. The advanced IDS for IIoT uses machine learning and artificial intelligence to raise the detection accuracy and bring down the number of false positives, thus, very cleverly adapting to the always changing threat environment [9]. Studies show that the installation of intrusion detection systems in Industrial Internet of Things

environments alone can lead to a reduction in the number of cyber-attacks by as much as 60%. Moreover, the surveillance and learning aspects of the IDS ensure that the industrial systems are always protected against new and developing threats. Continuing research and development in IDS technologies are necessary for keeping up with the changing landscape of cyber threats. The objective of this research is to analyze the current issues, innovations, and the AI-driven IDS solutions in the IIoT. It also provides a detailed review of the ways AI methods enhance IIoT security, highlights the restrictions, and recommends future research and implementation. The next parts of this survey are structured as follows: Section 2 investigates the configuration of the Industrial Internet of Things (IIoT). Section 3 gives a recap of Intrusion Detection Systems. Section 4 centers around the current work on AI-driven IoT IDS. Section 5 outlines some of the remaining major research challenges, while section 6 wraps up the study and outlines possible paths for future exploration.

II. BACKGROUND

A. The IIOT System Architecture

The IIoT can be defined as the networked integration of devices, sensors, and industrial systems. This technology has the potential to gather and analyze data which would consequently benefit the industrial sector through increased efficiency. The IIoT can be considered a form of IoT that incorporates cloud and front-end computing to enhance automation. Security in IIoT has recently become a subject of much interest, reminiscent of the situation surrounding IoT [10]. The objectives of IIoT security solutions include protecting devices and ensuring secure data transmission by making use of new programs and techniques. IIoT is a blessing for the industries as it allows for the merging of the physical and virtual worlds which leads to the enhancement of the operational efficiencies [11]. The Industrial Internet of Things (IIoT) has various protocols available for connecting to the Internet. The IIoT architecture is shown in Figure 1.

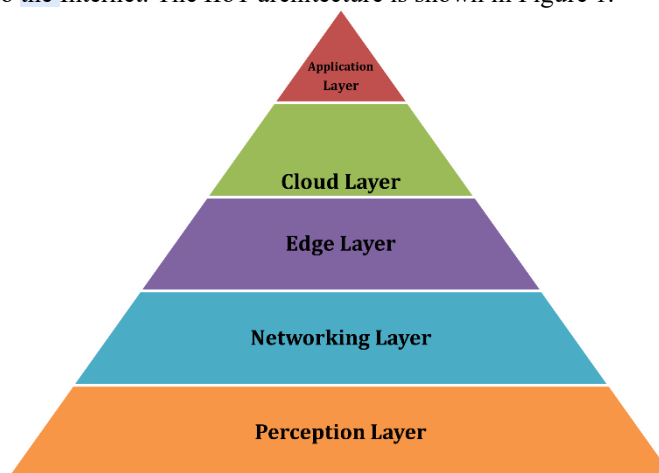


Fig. 1 Architecture of IIoT system

It consists of five layers: perception, networking, Edge or Fog Layer, cloud and application. The tangible components are addressed in the initial stratum [12]. The second layer encompasses communication protocols that enable the transfer of data to the third layer, which is optional but

necessary, does calculations near the devices to cut down on latency. This layer is used for pre-processing data (filtering, compression, and finding anomalies) to lower the strain on the cloud and speed up response times. The fourth layer provides comprehensive analytics, data storage, and scalability. The application layer is the top layer, where processed data is utilized to make industrial applications. A feature. All tiers of the IIoT are vulnerable to attacks and intruders. The major cybersecurity threats comprised DoS, DDoS as well as deception and spamming, while MiTM assaults were carried out sporadically. Many corporations make use of IDSs in order to pinpoint malevolent attacks and provide protection to the nodes and networks of IIoT [13].

Intrusion Detection Systems

Data integrity, confidentiality, and security in networks are greatly influenced by IDS. Therefore, it plays a major role in IIoT security. The main functions of an IDS include not only the very important detection but also preventing, responding to, and reporting the harmful acts in IIoT networks. It is possible to categorize and place the IDSs at any level of system architecture. Network monitoring employs packet data collection [14]. Cybercriminals often resort to using either malware or network packet analysis to retrieve sensitive information. Attacks could be directed at either the server that processes all network transactions or the workstation that is engaging in network activity. There might be opportunities for the exploitation of system vulnerabilities. Indeed, machine learning together with deep learning form the basis for the advanced intrusion detection systems that can very accurately identify network threats [15]. Data for intrusion analysis can come from multiple sources. Depending on the information sources, IDSs can be either host-based or network-based. The identification of intrusive and malicious activities will be through the examination of the infrastructure-specific technique in the case of the IDS [16]. In Figure 2, the analytical method indicates that the IDSs fall into the categories of anomaly-based and misuse-based.

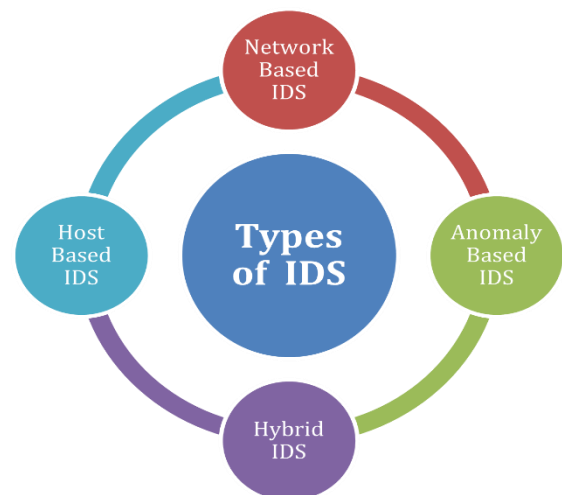


Fig. 2. IDS taxonomy system

III. RELATED WORK

Many frameworks and models have been proposed for identifying IIoT categories at the levels of device, network,

and database. The next part of the paper is about discussing the different solutions where machine learning and deep learning techniques have been used.

O. Almomani et al. [17] An IDS framework has been implemented with the different classifiers. The feature selection process involved three algorithms such as Genetic Algorithm (GA) using UNSW-NB15 dataset. The performance of GA-J48, GWO-J48, and FFA-J48 was measured in terms of accuracy where they got 86.874%, 85.676%, and 86.037% respectively. Likewise, accuracy ratings of 86.387%, 84.485%, and 85.429% were recorded for GA-SVM, GWO-SVM, and FFA-SVM, respectively. Therefore, the combination of J48 and SVM algorithms rendered remarkable results.

Z. Chen et al. [18] utilized different XGBoost thresholds as a method for feature selection. The proposed model was tailored to feature selection and loss function thresholds that would be the best for our model. The MIX_LSTM model outperforms DL, ML, and even the finest algorithms at detecting rare Industrial Internet of Things risks.

A. Yazdinejad et al. [19] deep learning model based on Long Short-Term Memory (LSTM) and Auto-Encoder (AE) in ensemble is applied to detect abnormal behavior for cyber threat hunting in IIoT. The model employs unbalanced IIoT datasets from Gas Pipeline (GP) and Secure Water Treatment (SWaT). Traditional machine learning classifiers such as Random Forest give lower accuracy than proposed model.

S. Tharewal et al. [20] introduced a deep reinforcement learning method was presented which could be regarded as a very efficient near-end strategy optimization technique for the IDS. The DRL-IDS system uses LightGBM to quickly and effectively extract the most pertinent feature set from the IoT data and deploys deep learning techniques for carrying out the intrusion detection process. The proposed solution has been able to detect 99 percent of the IoT network attacks using some publicly available dataset. Moreover, it has a false acceptance rate of 0.9%.

T. Gaber et al. [21] introduced a new approach that featured the use of PSO and BA for selecting the characteristics, while Random Forest (RF) was the classifier used for IIoT network traffic classification. The model was tested on the dataset WUSTL-IIOT-2021. The objective was to select the best one by comparing the results of the two feature selection methods. RF+BA was the classifier that performed the best.

H. V. Krishna and B.Samatha [22] proposed method that employs ensemble classifiers for the purpose of detecting attacks. IIoT networks are constantly changing and interconnected. Hence, the proposed system is capable to cope with them. The enhanced Adaboost algorithm works on the principle of weighted error and by giving priority to misclassified occurrences, it not only increases the accuracy of the RF classifier but also gives rise to the formation of diverse and flexible clusters. The experimental results confirm that it is capable of rapidly detecting and preventing intrusions.

Z.Huma et al. [23] introduced the utilization of HDRaNNs for pinpointing IIoT network breaches has been researched extensively. The architecture is composed of deep random

neural networks and multilayer perceptrons that apply dropout regularization. The HDRaNN model classified the DS2OS and UNSW-NB15 datasets with respective accuracies of 98% and 99% after having identified 16 types of cyberattacks.

IV. CHALLENGES FACED BY IDS IN THE INDUSTRIAL INTERNET OF THINGS

The links of IIoT enable machine-to-machine communication in diverse fields [23]. Industrial IoT applications are very important but are sometimes unreliable due to the differences in technology, constantly changing environments, and the large amount of data generated. IIoT has its advantages as well as disadvantages. The ability to quickly analyze large volumes of data from Industrial Internet of Things (IIoT) devices requires many critical features. Examples are proper device maintenance, network security, data confidentiality, access control, machine and human error. Not tackling these problems may have a negative impact on the performance of the industry.

The IIoT brings along a new set of operational, environmental, and architectural limitations that make the Intrusion Detection System design, deployment, and use very difficult. The main IDS issues are shown in Figure 3. A list of the main problems is presented below [24][25]:

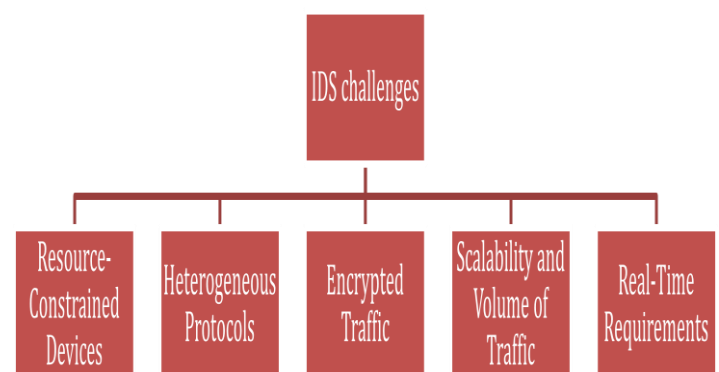


Fig. 3. IDS challenges

- **Devices with limited resources:** Many IIoT endpoints, such as sensors, actuators, PLCs, and RTUs, don't have a lot of processing power, memory, battery life, or storage. Traditional IDS that use signatures or deep packet inspection need a lot of processing power and memory, thus they can't be installed directly on edge devices. We need lightweight models for detecting anomalies, but getting good accuracy with less resources is still hard.
- **Different Protocols and Old Systems:** IIoT settings employ both old and new industrial protocols. Some examples of new protocols include MQTT, OPC UA, CoAP, and DDS. Some examples of old protocols are Modbus/TCP, PROFINET, EtherNet/IP, S7comm, DNP3, and IEC 61850. Many old

protocols don't have encryption, authentication, or adequate checks for message integrity built in. Keeping up-to-date signature databases for dozens of proprietary and poorly described protocols is quite hard.

- Requirements for Real-Time and Deterministic: Industrial control systems need communication that is both low-latency and predictable, usually in the millisecond range. Deep packet inspection or complicated behavioral analysis might cause delays or jitters that are not acceptable. IDS must work in a way that is completely non-intrusive (passive) or with assured limited latency.
- How much traffic it can handle and how big it can become: IIoT networks may include tens or hundreds of thousands of devices that provide telemetry and control data all the time. Centralized IDS systems might become sites of failure and bottlenecks. We need distributed and hierarchical IDS designs, but they make things more complicated and take more time to administer.
- Traffic that is encrypted: More people are using TLS/DTLS, IPsec, and encryption that is particular to a protocol (like OPC UA with message security). This makes payload-based detection useless. Conventional deep packet inspection fails unless expensive man-in-the-middle decryption proxies are used, which are often deemed undesirable in operational technology settings owing to certification and safety concerns.

V. FUTURE DIRECTIONS FOR IIOT

The use of AI technology in IIoT ecosystem security through intrusion detection is still a long way to go, though it has been a gradual process. The challenges that need to be tackled further are crucial in developing secure, scalable and reliable IDS products for the advanced industrial systems [26][27].

- IDS with Edge Intelligence and Lightweight : The future exploration of this area should focus on the development of lightweight and computationally efficient AI models for IIoT edge devices. To achieve real-time intrusion detection with low latency, energy, and communication overhead, model pruning, quantization, and TinyML will be utilized.
- Federated, Private Learning: Federated learning along with decentralized AI frameworks will facilitate the overcoming of data privacy and bandwidth problems. These techniques will allow the remote IIoT nodes to train the IDS models without sharing the raw data, hence increasing the scalability while at the same time safeguarding the industrial data that is sensitive [28].
- Explainable, reliable AI : IIoT systems are critical to safety, thus the future IDS must incorporate XAI. The detection decision explanations that are transparent and comprehensible not only enhance the operator's confidence and incident response but also give industrial cybersecurity compliance a boost.
- Self-learning and adaptive detection mechanisms: Traditional IDS are not equipped to handle zero-day and evolving attacks. The future AI-driven IDS must apply continuous learning and self-adaptive techniques to make the detection models update according to new traffic patterns without losing stability or performance [29].

- Integrating Digital Twins :The combination of IDS and digital twin technologies results in proactive security. Digital twins can simulate cyberattacks, generate authentic training data, and validate security measures before they are applied in the industrial systems [30].

VI. CONCLUSION

The swift expansion of the IIoT opens new avenues for innovation and higher efficiency in industrial operations like never before. Still, the larger area of possible attacks and the vital importance of industrial systems also create a lot of difficulties in terms of cybersecurity. The integration of AI in IDS enhances their efficacy and guarantees their capacity to address the specific challenges of IIoT, including device heterogeneity and real-time operational demands. In this survey, we analyzed ML and DL methods employed in the development of Intrusion Detection Systems (IDS) over the past five years, specifically from 2020 to 2026. Although progress has been achieved, numerous areas still require further research and development. Enhancing the robustness of AI-driven IDS against advanced assaults, improving their scalability for extensive IIoT deployments, and ensuring their reliability across diverse industrial environments are essential areas that warrant further investigation. By progressing these technologies and overcoming their existing limitations, we will be able to greatly enhance the security and reliability of industrial systems, which in turn will give a guarantee to the future safe and efficient operation of IIoT infrastructures.

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